

Digital Forensics Incident Report

Task: Week 2 - Incident Simulation & Evidence Collection

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Date: May 1, 2026

1.Executive Summary

This report documents the simulation of a phishing attack and the subsequent forensic collection of digital artifacts. The simulation involved a banking-themed social engineering attack targeting a Windows 10 victim. Evidence collected includes network traffic logs, system activity, and exfiltrated credentials.

2. Incident Simulation

The attack was orchestrated using Gophish on a Kali Linux machine. A 'National Bank of Egypt' phishing template was used to deceive the victim into submitting their Banking credentials.

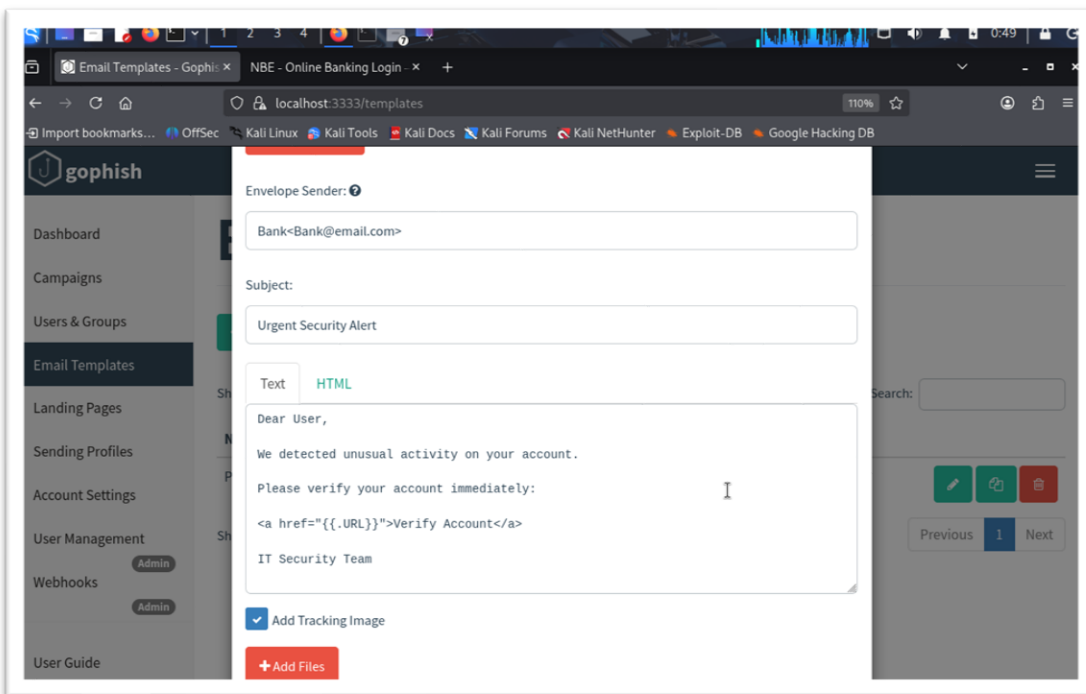
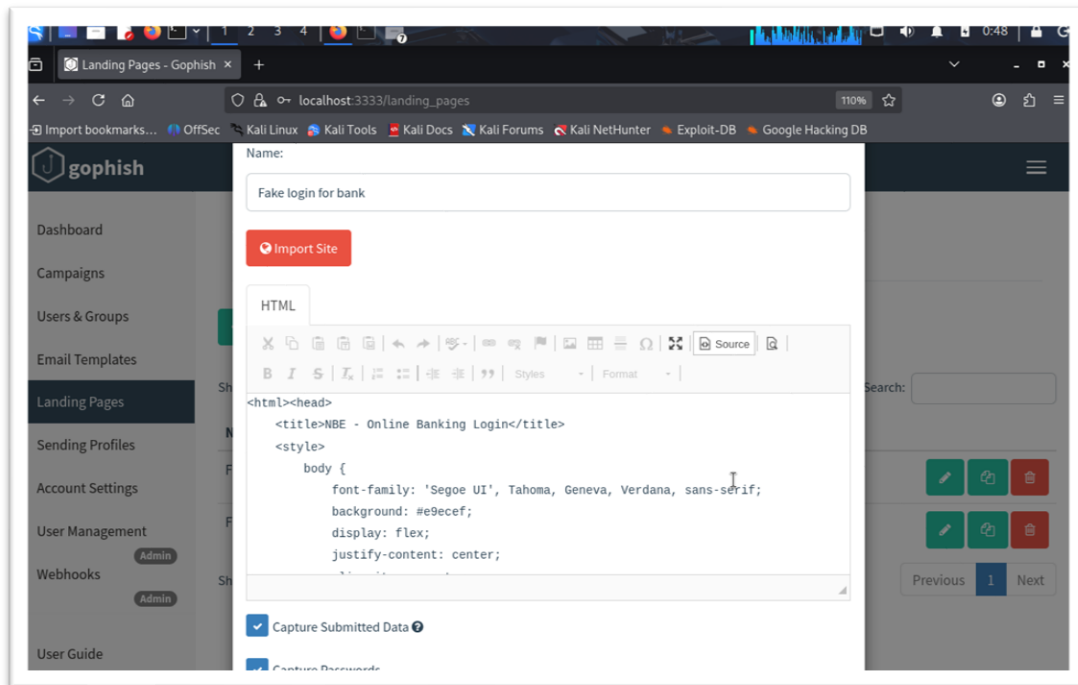
2.1 Environment Details

Attacker Machine: Kali Linux (IP: 192.168.56.101) running Gophish.

Victim Machine: Windows Lab (IP: 192.168.56.103).

Analysis Tools: Wireshark, Sysmon (v15.0+), MXToolbox.

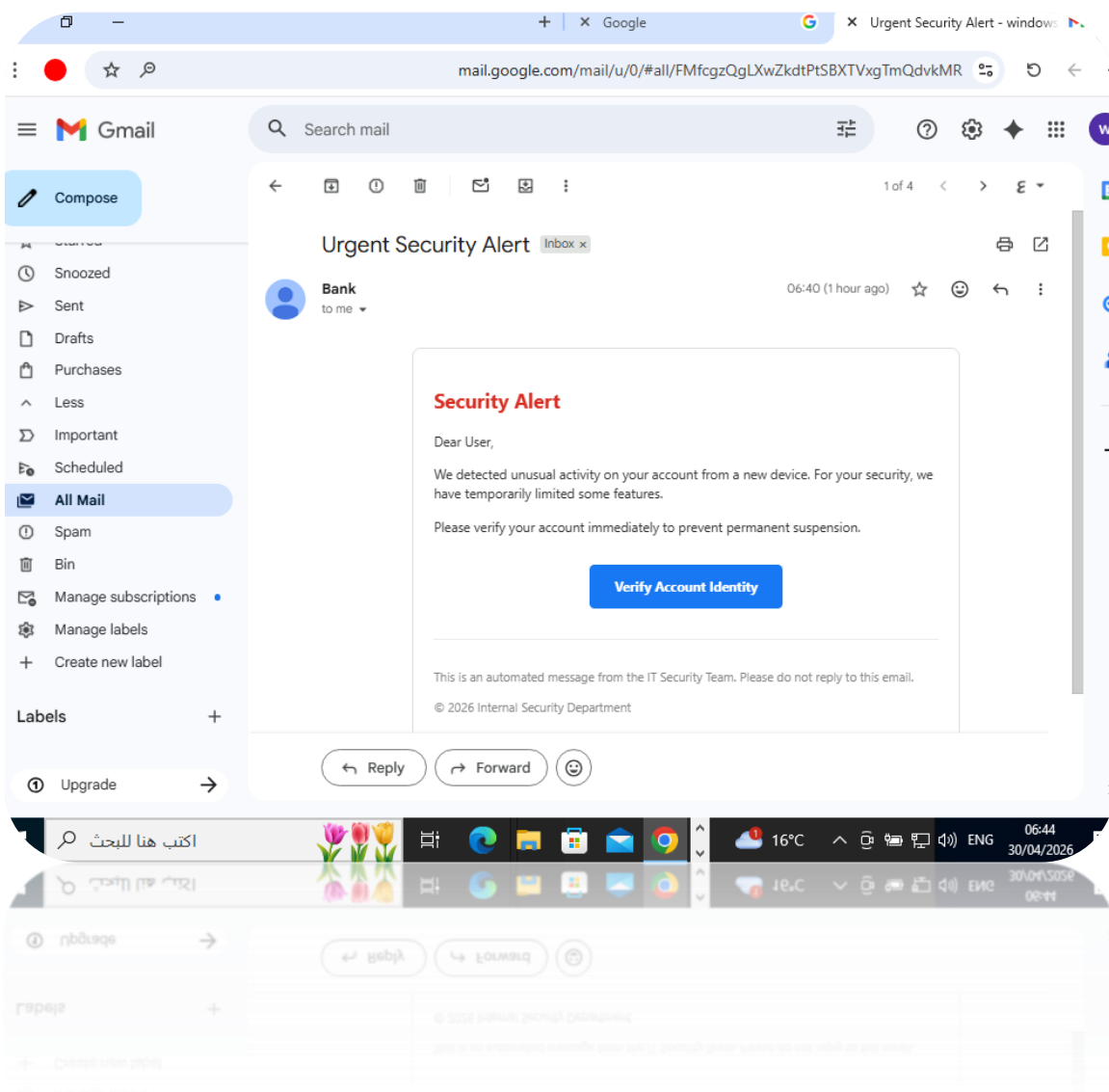
2.2 Phase 1 : Attack prepration



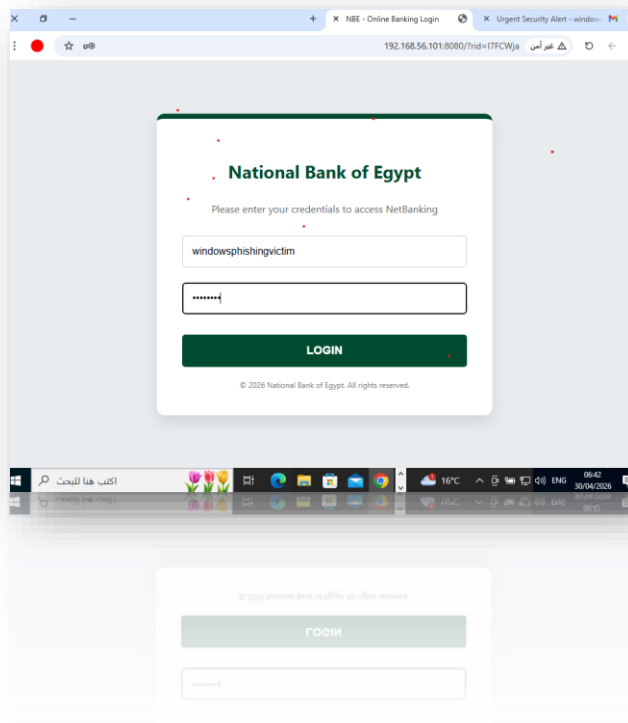
2.3 Phase 2 : Attack simulation on kali for Testing

- Click here to watch the Attack Simulation Video: [Attack Simulation Video](#)

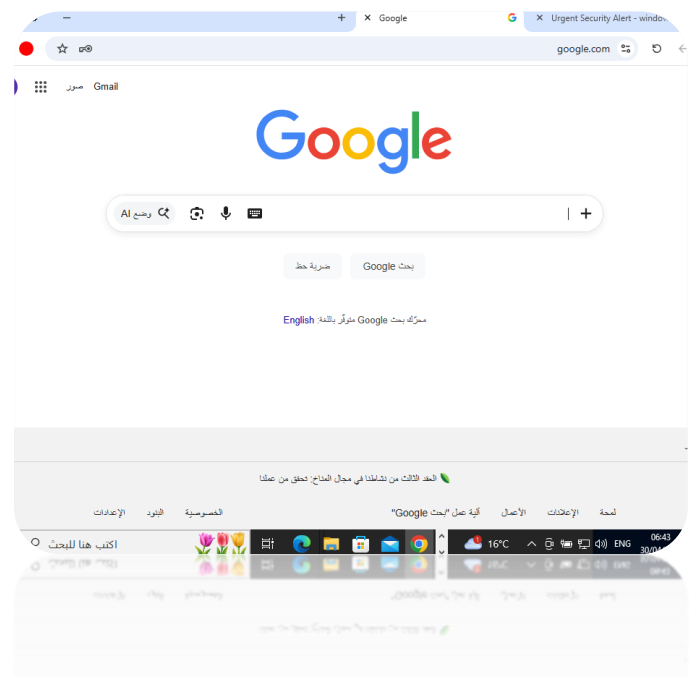
2.4 Phase 3 : Attack Simulation on windows



Fake Login Page:



victim redirected to Google.com:



3. Incident Timeline

Time (April 29, 2026)	Event	Source
11:40:11 PM	Phishing Email Sent	Gophish
11:41:05 PM	Victim Clicked Link	Windows 10 / Chrome
11:43:18 PM	Credentials Submitted	Landing Page
11:43:19 PM	Captured POST Request	Attacker Terminal
May 1 - 02:37 AM	Infrastructure Scan	MXToolbox

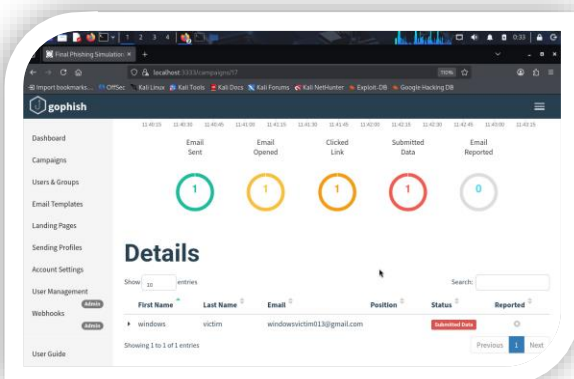
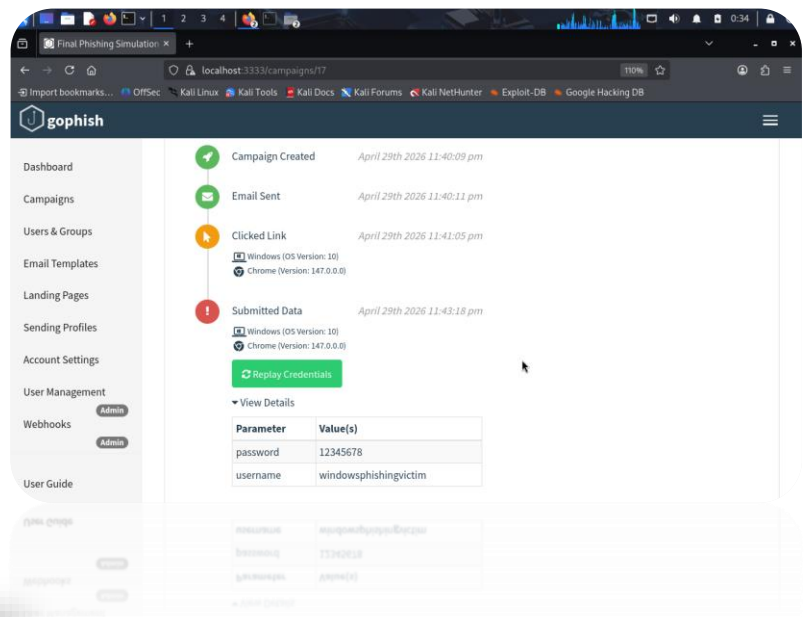
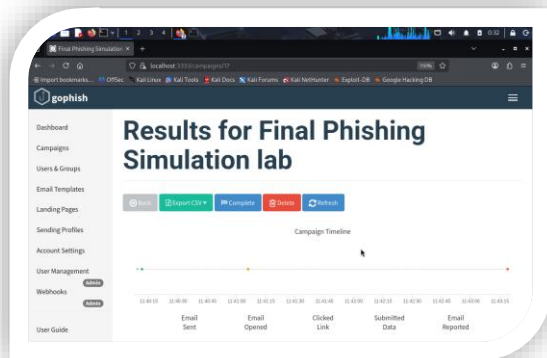
4. Evidence Collection & Analysis

4.1 Captured Credentials

The phishing server successfully captured the following plain-text data:

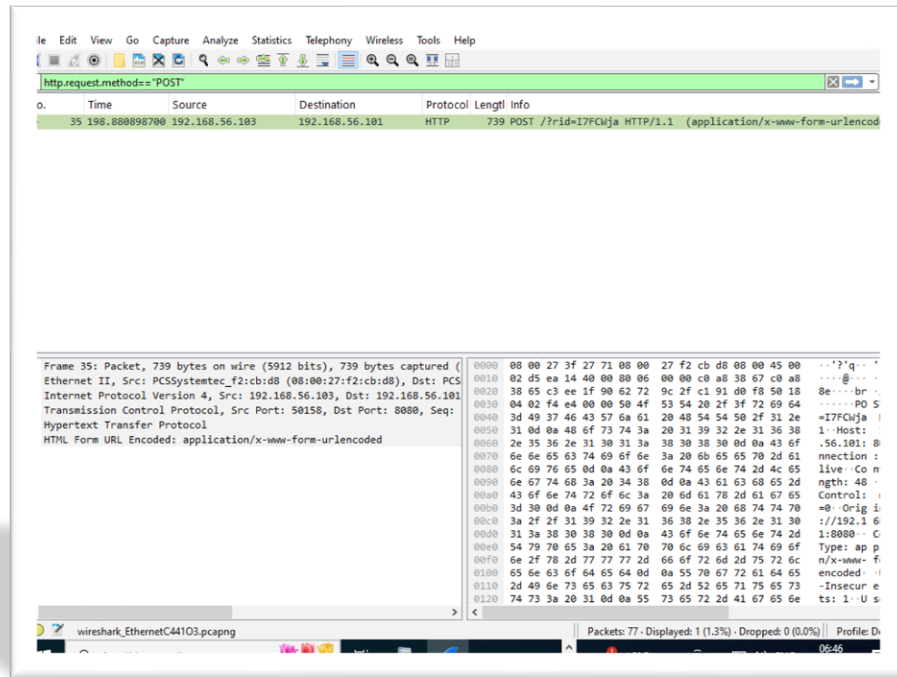
- Username: windowsphishingvictim

- Password: 12345678

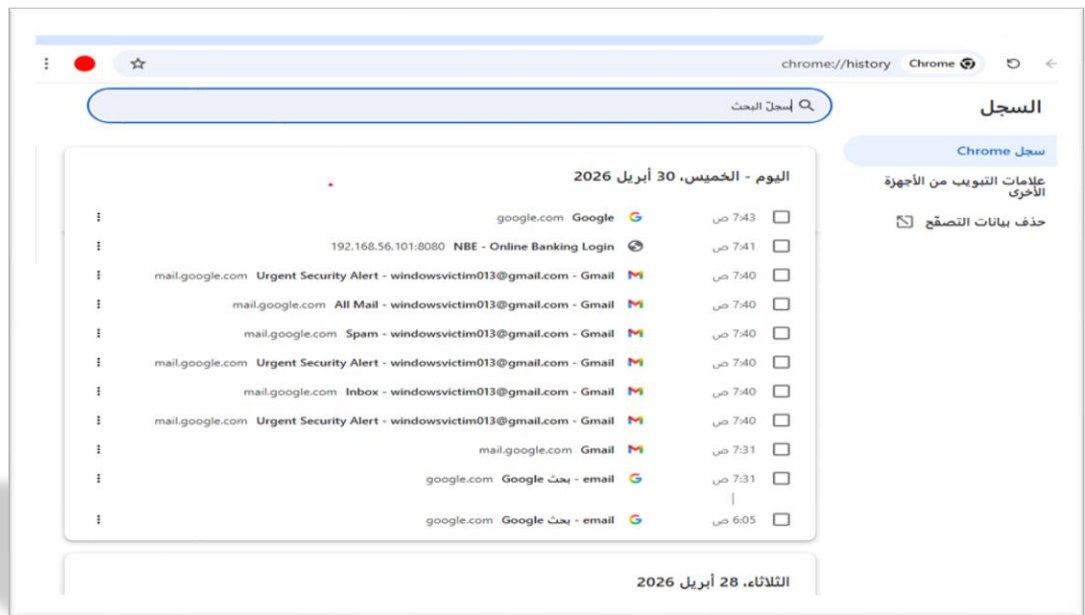


4.2 Network Traffic (Wireshark)

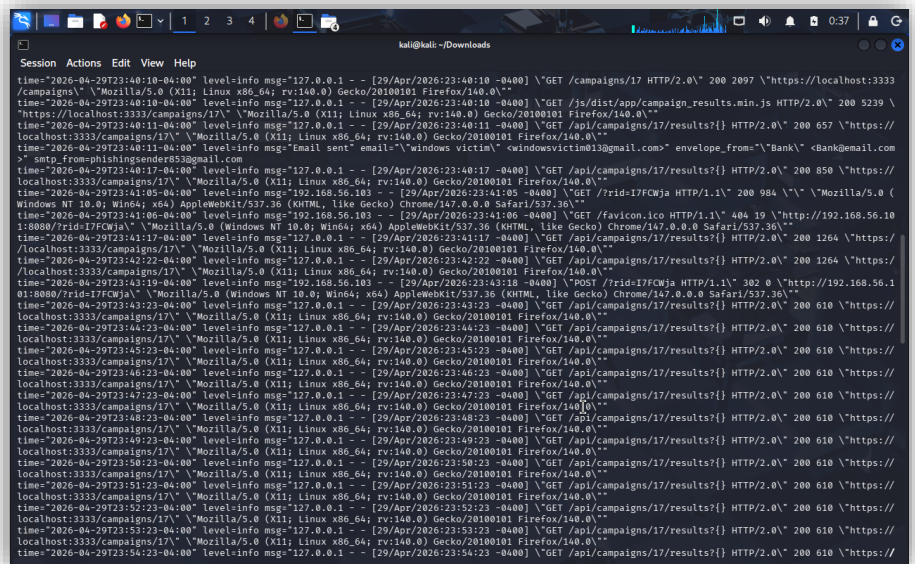
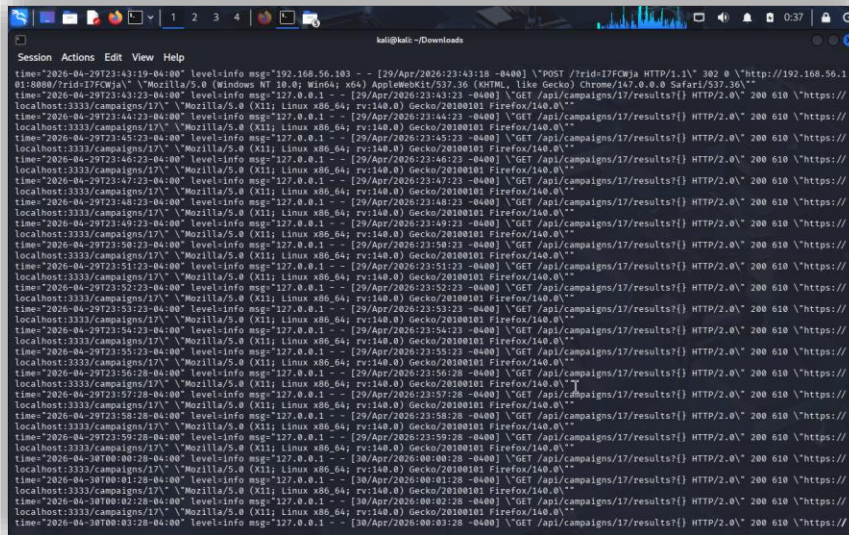
The evidence shows a clear HTTP POST communication between 192.168.56.103 (Victim) and 192.168.56.101 (Attacker). Logs confirm the data transfer occurred at port 8080.



4.3 Browser history on windows (victim)



4.4 Attacker-Side Web Server Logs (Kali Linux)



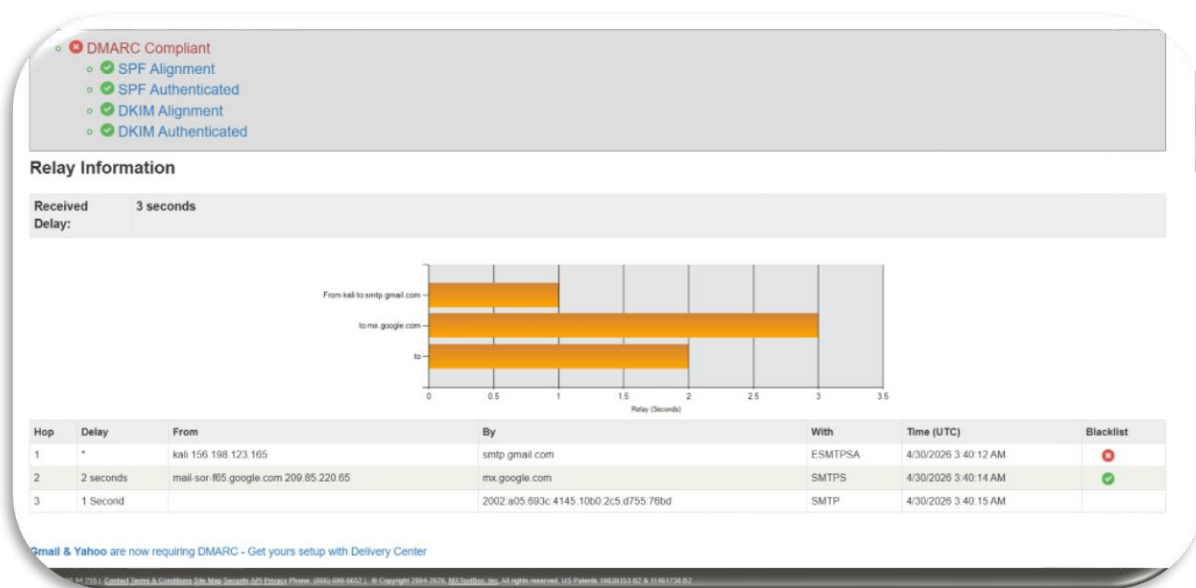
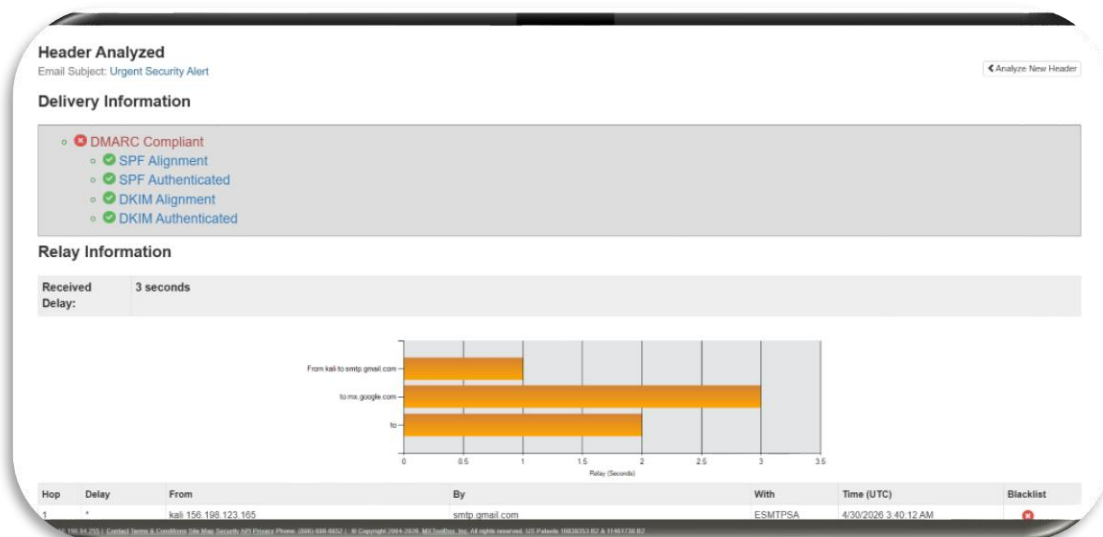
These logs show the victim (**192.168.56.103**) connecting to the attacker's server (**192.168.56.101**). The **POST request** at 11:43:18 PM is the most important evidence, as it proves the victim sent their stolen username and password directly to the attacker.

4.5 Email Authentication Analysis

4.5.1 Header analysis

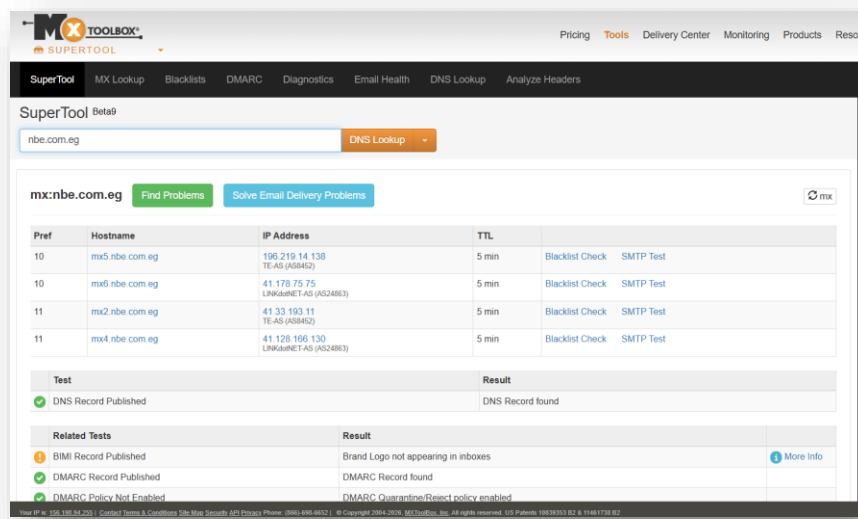
```
Delivered-To: windowsvictim013@gmail.com
Received: by 2002:a05:693c:4145:10b0:2c5:d755:76bd with SMTP id w5-nlcsip657460dya; Wed, 29 Apr 2026 20:40:15 -0700 (PDT)
X-Received: by 2002:a05:600c:4e8c:b0:48a:6315:da26 with SMTP id 5b1f17b1804b1-48a84482a42mr19679635e9.26.1777520414833; Wed, 29 Apr 2026 20:40:14 -0700 (PDT)
ARC-Seal: i=1; a=rsa-sha256; t=1777520414; cv=none; d=google.com; s=arc-20240605; b=CuH70+IY20mlK9TnHfZ7nQUPgk2eDK0h8Lk57VpVxTQ5NSxv0j0k+zyH70CPiWdPa
VP0zGA/nkZZz9Jazp7P6U9SE0y0w1z6H05vnmY8qDGZaF/z3ufiV9MyGuCiqerwH aku0a534FwcihKwDccfUTY1eDeCqHvRLjGa5T/3g52awQFy0d0T+XU401tb0Uur/X1FM
xzC037AGihg4sCXIAEIEtXtJZenou8Wm169Iqks23IV9cJH3w4P1R3q+CDgS90pYpD LZ74UUXVFMqR8t0gZx5ZzTJmJ986hRGJlHpdw8TwhukGa8F5r1+sBXL7bg3drul0s sptQ==
ARC-Message-Signature: i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605; h=message-id:to:subject:date:mime-version:from:dkim-
signature; bh=93gTveQZ0T5CBRDGKdksR7L73eivQmno4Bm483Ww+7s=; fh=n1soG4w0C8DCsaUl8/Z57xeInfvFq/FyJvM80pAl+;
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zz/NyEE782+hbvYd1qXN5eeQzmtLET+y6MPduqaASpDVtn/DLeopqX5/BCNN2eQOMA I16g==; dara=google.com
ARC-Authentication-Results: i=1; mx.google.com; dkim=pass header.i=@gmail.com header.s=20251104 header.b=pxcvV80e; spf=pass (google.com: domain of
phishingsender853@gmail.com designates 209.85.220.65 as permitted sender) smtp.mailfrom=phishingsender853@gmail.com; dmarc=pass (p=NONE sp=QUARANTINE
dis=NONE) header.from=gmail.com; dara=pass header.i=@gmail.com
Return-Path: <phishingsender853@gmail.com>
Received: from mail-sor-f65.google.com (mail-sor-f65.google.com. [209.85.220.65]) by mx.google.com with SMTPS id ffacd0b85a97d-
4494f8a3f63sor54013f8f.15.2026.04.29.20.40.14 for <windowsvictim013@gmail.com> (Google Transport Security); Wed, 29 Apr 2026 20:40:14 -0700 (PDT)
Received-SPF: pass (google.com: domain of phishingsender853@gmail.com designates 209.85.220.65 as permitted sender) client-ip=209.85.220.65;
Authentication-Results: mx.google.com; dkim=pass header.i=@gmail.com header.s=20251104 header.b=pxcvV80e; spf=pass (google.com: domain of
phishingsender853@gmail.com designates 209.85.220.65 as permitted sender) smtp.mailfrom=phishingsender853@gmail.com; dmarc=pass (p=NONE sp=QUARANTINE
dis=NONE) header.from=gmail.com; dara=pass header.i=@gmail.com
DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=gmail.com; s=20251104; t=1777520414; x=1778125214; dara=google.com; h=message-
id:to:subject:date:mime-version:from:from:to:cc:subject:date:message-id:reply-to; bh=93gTveQZ0T5CBRDGKdksR7L73eivQmno4Bm483Ww+7s=;
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3La79HfFyG8cso+FL9LzeVwJm1f8YBz+E0yz711F3VHUC0mXyl+F0Z0g5Th7rZ588 my+rrG/Vt4yD0RM70awQE3FHjzTGL0S9LLWISJ2U55j1Z2w6X0INUWwP3s5j+Y15/Shv
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X-Google-DKIM-Signature: v=1; a=rsa-sha256; c=relaxed/relaxed; d=1e100.net; s=20251104; t=1777520414; x=1778125214; h=message-id:to:subject:date:mime-
version:from:from:cc:subject:date:message-id:reply-to; bh=93gTveQZ0T5CBRDGKdksR7L73eivQmno4Bm483Ww+7s=;
b=PTZT0X0B8jM20XV0h/LmU0UesqpsI/GqrdiF90ubJp+8fSGtoI4f8xLZnrIK/X 2500VASpawK0q01Ry3ryuJjKenze5EY78VxmvgmK7MTZTVv07pqTbFPULdzfjjzr
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STGAdppd15ZulEmI26p/051b0Q9THjbeASuocYf9QevQvX80P6H93Eq/pfw1n3Z38JA R2aw==
X-Gm-Message-State: A0JubY7Z7/RcT8p6BnyKGSx0EYvH5/hmoxHf13VgE26QU+1YxH0dGI N1106VA1Buf+wfS1qk/WamCoJ6y8MSx/m5AH1lUigkUWt/CTMvRdKJTAz8X2w+13+k=
X-Gm-Gg: AebDietvcI0wPTB0DgnpxyAA6yE1480MEN6+535CKVYsEybaes56J5ICVKZhuqcd
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X-Received: by 2002:a05:6000:2585:b0:43b:498f:dceb with SMTP id ffacd0b85a97d-4493c37ef0cmr1659818f8f.9.1777520413576; Wed, 29 Apr 2026 20:40:13 -0700 (PDT)
Return-Path: <phishingsender853@gmail.com>
Received: from kail (f156.198.123.165) by smtp.gmail.com with ESMTPS id ffacd0b85a97d-447b4216ed6sm10087097f8f.16.2026.04.29.20.40.12 for
```

Forensic examination of the **Authentication-Results** header reveals that while the email passed SPF and DKIM checks, it did so under the domain *gmail.com* (Sender: *phishingsender853@gmail.com*). This confirms a **Business Email Compromise (BEC)** style attack where a legitimate service provider is abused to deliver malicious content. The headers clearly show the email originated from a personal Gmail account rather than the official domain of the financial institution it attempted to impersonate.



Analysis of phishing email headers to investigate sender authenticity, mail relay chain, and email security controls (SPF, DKIM, DMARC)

4.5.2 Infrastructure Verification



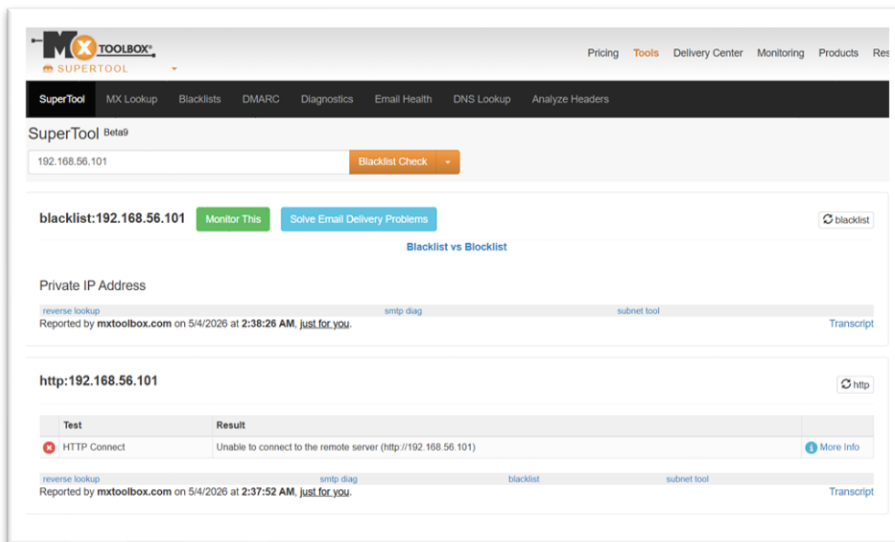
The screenshot shows the MXToolbox SuperTool interface for a DNS Lookup of `nbe.com.eg`. The tool displays a table of mail exchangers (MX records) for the domain. Below the table, it shows the results of various tests, including DNS Record Published, BIMI Record Published, DMARC Record Published, and DMARC Policy Not Enabled.

Pref	Hostname	IP Address	TTL	Blacklist Check	SMTP Test
10	mx5.nbe.com.eg	199.219.14.138 TE-AS (AS8462)	5 min	Blacklist Check	SMTP Test
10	mx6.nbe.com.eg	41.178.75.75 LINKARNET-AS (AS24863)	5 min	Blacklist Check	SMTP Test
11	mx2.nbe.com.eg	41.33.193.11 TE-AS (AS8462)	5 min	Blacklist Check	SMTP Test
11	mx4.nbe.com.eg	41.128.199.130 LINKARNET-AS (AS24863)	5 min	Blacklist Check	SMTP Test

Test	Result
✓ DNS Record Published	DNS Record found

Related Tests	Result	More Info
⚠ BIMI Record Published	Brand Logo not appearing in inboxes	More Info
✓ DMARC Record Published	DMARC Record found	
✓ DMARC Policy Not Enabled	DMARC Quarantine/Reject policy enabled	

A DNS lookup via MXToolbox for the legitimate domain *nbe.com.eg* shows a completely different set of authorized mail servers. This contrast proves that the attacker's infrastructure (**192.168.56.101**) has zero technical affiliation with the actual National Bank of Egypt, confirming it is an unauthorized, malicious source.



The screenshot shows the MXToolbox SuperTool interface for a Blacklist Check of the IP address `192.168.56.101`. The tool displays the results of various tests, including HTTP Connect, which failed due to an inability to connect to the remote server.

blacklist:192.168.56.101 [Monitor This](#) [Solve Email Delivery Problems](#)

Blacklist vs Blocklist

Private IP Address

[reverse lookup](#) [smtp diag](#) [subnet tool](#) [Transcript](#)

Reported by mxtoolbox.com on 5/4/2026 at 2:38:26 AM. [Just for you.](#)

Test	Result	More Info
⚠ HTTP Connect	Unable to connect to the remote server (http://192.168.56.101)	More Info

[reverse lookup](#) [smtp diag](#) [blacklist](#) [subnet tool](#) [Transcript](#)

Reported by mxtoolbox.com on 5/4/2026 at 2:37:52 AM. [Just for you.](#)

External scans confirmed the IP was a Private/Internal address. The connection failure noted in later scans confirms the server was decommissioned post-attack.

5. Risk Assessment

The attacker specifically chose to clone the National Bank of Egypt (NBE) portal, which is a critical piece of evidence regarding the attack's severity. By targeting the largest state-owned bank in Egypt, the attacker aimed to exploit a massive user base. This '**Brand Impersonation**' is a high-risk tactic because it leverages the high level of trust users place in national financial institutions. From a forensic perspective, the use of NBE branding in the phishing template and the landing page indicates a targeted social engineering effort (Spear Phishing) designed to cause maximum financial impact and identity theft.

6. Deliverables

The following digital artifacts are attached and linked for review:

- **Raw Wireshark Logs (.pcapng):**

- Post Method on Wireshark

- Wireshark Logs Filtered with ip address of attacker

- **Sysmon Event Logs (.evtx):**

- Sysmon Logs Filtered by ID 3

- Sysmon Logs Filtered By ID 1